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Author(s): Borislav Bojanov

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NOTES

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Elementary Proof of the Remez Inequality

Borislav Bojanov

This note is concerned with the Tchebycheff polynomials $T_n(x)$. As well known they can be presented on $[-1, 1]$ by the expression

$$T_n(x) = \cos(n \arccos x).$$

The famous Russian mathematician Pafnutii Lvovich Tchebycheff (1821–1894) introduced $T_n(x)$ as the polynomial of least uniform norm on $[-1, 1]$ amid the polynomials of degree n with fixed leading coefficient.

The Tchebycheff polynomials appear prominently in various extremal problems posed in π_n (the set of all polynomials of degree n). An illuminating example is the classical Markov inequality, which shows that

$$\|p^{(k)}\| \leq \|T_n^{(k)}\|, \quad k = 0, \dots, n,$$

for each $p \in \pi_n$ such that

$$\|p\| := \max\{|p(x)| : x \in [-1, 1]\} \leq 1.$$

The proof of this and many other remarkable properties of T_n can be found in the recent book of Rivlin [4].

It has been mentioned already by Tchebycheff that T_n is the fastest growing polynomial outside $[-1, 1]$. In other words,

$$\max\{|p(\xi)| : p \in \pi_n, \|p\| \leq 1\} = T_n(\xi)$$

for each $|\xi| \geq 1$. This observation provokes the following question: How large can a polynomial be given that it is constrained to be “small” on a substantial portion of its domain? Make the problem more precise as follows.

Let σ be an arbitrary fixed positive number. For every $p \in \pi_n$ define the set

$$M(p) := \{x \in [-1, 1 + \sigma] : |p(x)| \leq 1\}.$$

Clearly $M(p)$ consists of mutually disjoint closed subintervals. Let $|M(p)|$ be the measure of $M(p)$, i.e., $|M(p)|$ is the total length of these subintervals. Denote

$$\pi_n(\sigma) := \{p \in \pi_n : |M(p)| \geq 2\}.$$

The problem is to characterize the polynomial p^* from $\pi_n(\sigma)$ which has a maximal uniform norm over $[-1, 1 + \sigma]$.

Evidently, the Tchebycheff polynomial $T_n(x)$ belongs to $\pi_n(\sigma)$ for each $\sigma > 0$ since $|T_n(x)| \leq 1$ on $[-1, 1]$ and $|T_n(x)| > 1$ for $|x| > 1$. In 1936 Remez [1]

established the following

$$\sup_{p \in \pi_n(\sigma)} \|p\|_\infty = \|T_n\|_\infty, \tag{1}$$

where the supremum norm is over $[-1, 1 + \sigma]$. Of course $\|T_n\|_\infty = T_n(1 + \sigma)$. The proof of (1) can be seen also in the book of Freud [2]. A simpler approach was found recently by Erdelyi [3]. We demonstrate here a short, elementary proof.

The proof: Note that for any fixed $x \in [-1, 1 + \sigma]$ the quantity

$$\mu(x) := \sup\{|p(x)| : p \in \pi_n(\sigma)\}$$

is attained for some polynomial from $\pi_n(\sigma)$. We shall show first that $\mu(x) \leq \mu(1 + \sigma)$ for each $x \in [-1, 1 + \sigma]$. Indeed, let x be an interior point of $[-1, 1 + \sigma]$ and let p be the extremal polynomial for this point, i.e., $p \in \pi_n(\sigma)$ and $|p(x)| = \mu(x)$. Introduce the polynomials

$$p_1(x) := p(\alpha(x)), \quad p_2(x) := p(\beta(x)),$$

where $\alpha: [-1, 1 + \sigma] \rightarrow [-1, x]$ and $\beta: [-1, 1 + \sigma] \rightarrow [x, 1 + \sigma]$ are the linear transformations. Let M_1 and M_2 be the parts of $M(p)$ situated in $I_1 := [-1, x]$ and $I_2 := [x, 1 + \sigma]$, respectively. Assuming that $|M_i| < \lambda |I_i|$ for $i = 1, 2$ and $\lambda = 2/(2 + \sigma)$ we would get $|M| = |M_1 + M_2| < \lambda |I_1 + I_2| = \lambda(2 + \sigma) = 2$, a contradiction. Therefore $|M_i|/|I_i| \geq \lambda$ at least for one i , say for $i = 1$. Then $|M(p_1)| \geq 2$ and hence $p_1 \in \pi_n(\sigma)$. This yields

$$\mu(x) = |p(x)| = |p_1(1 + \sigma)| \leq \mu(1 + \sigma).$$

Therefore the Remez inequality will be proved if we show that

$$|p(1 + \sigma)| \leq T_n(1 + \sigma) \quad \text{for each } p \in \pi_n(\sigma).$$

In order to show this, denote by $-1 = \eta_0 < \eta_1 < \dots < \eta_n = 1$ the extremal points of T_n . We have

$$T_n(\eta_k) = (-1)^{n-k} \quad k = 0, \dots, n. \tag{2}$$

Let $x_0 < x_1 < \dots < x_n$ be the points of $M(p)$ which coincide with $\eta_0, \dots, \eta_n$ after we press $M(p)$ to the left, i.e., to the interval $[-1, M(p) - 1]$. By the Lagrange interpolation formula

$$|p(1 + \sigma)| \leq \sum_{k=0}^n \prod_{\substack{i=0 \\ i \neq k}}^n \frac{|1 + \sigma - x_i|}{|x_k - x_i|}$$

since $|p(x_i)| \leq 1$. Now taking into account the obvious inequalities $|1 + \sigma - x_i| \leq |1 + \sigma - \eta_i|$, $|x_k - x_i| \geq |\eta_k - \eta_i|$ and (2), we get

$$|p(1 + \sigma)| \leq \sum_{k=0}^n \prod_{\substack{i=0 \\ i \neq k}}^n \frac{|1 + \sigma - \eta_i|}{|\eta_k - \eta_i|} = T_n(1 + \sigma).$$

The proof is completed.

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2. G. Freud, *Orthogonal Polynomials*, Pergamon Press, Oxford, 1971.
3. T. Erdelyi, *Inequalities for generalized polynomials and their applications*, Ph.D. Thesis, University of South Carolina, 1989.
4. T. J. Rivlin, *Chebyshev Polynomials*, second edition, John Wiley & Sons, Inc., New York, 1990.

Department of Mathematics,
University of Sofia,
Boul. James Boucher 5,
1126 Sofia, BULGARIA

A Note on an Identity of Ramanujan

T. S. Nanjundiah

In a forthcoming paper [1], Berndt and Bhargava have supplied a proof of this eye-catching identity of Ramanujan found in his third notebook [3, p. 386]: if $ad = bc$, then

$$\begin{aligned} & 64\{(b+c+d)^6 - (a+c+d)^6 - (a+b+d)^6 + (a+b+c)^6 \\ & \quad + (a-d)^6 - (b-c)^6\} \\ & \times \{(b+c+d)^{10} - (a+c+d)^{10} - (a+b+d)^{10} \\ & \quad + (a+b+c)^{10} + (a-d)^{10} - (b-c)^{10}\} \\ & = 45\{(b+c+d)^8 - (a+c+d)^8 - (a+b+d)^8 \\ & \quad + (a+b+c)^8 + (a-d)^8 - (b-c)^8\}^2. \end{aligned}$$

It figures also in their expository article [2] featuring a selected group of Ramanujan's results. Unfortunately, they have missed its simple proof and so its genesis by not noticing that it is built from two sets of sums:

$$\begin{aligned} u_n &= \alpha_1^n + \beta_1^n + \gamma_1^n, & \alpha_1 &= b+c+d, & \beta_1 &= -(a+b+c), & \gamma_1 &= a-d, \\ v_n &= \alpha_2^n - \beta_2^n + \gamma_2^n, & \alpha_2 &= a+c+d, & \beta_2 &= -(a+b+d), & \gamma_2 &= b-c. \end{aligned}$$

By $\alpha_j + \beta_j + \gamma_j = 0$, the underlying problem is to compute

$$\omega_n = \alpha^n + \beta^n + \gamma^n,$$

where α , β and γ are the roots of the cubic

$$z^3 - pz + q = 0.$$